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Pectoralis Major Injuries in the National Football League

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Background: Pectoralis major (PM) injuries are rare, primarily occurring in males during athletic activity. In the current literature, these injuries have not been well described in National Football League (NFL) athletes.

Hypothesis: The incidence of PM injuries will be low in NFL athletes, with athletes missing significantly more time after injuries requiring operative management.

Study Design: Cohort study.

Level of Evidence: Level 4.

Methods: All documented PM injuries were retrospectively analyzed using the NFL Injury Surveillance System over a 15-season period. The data were analyzed by season, session, position, activity, and contact type at the time of injury. Additionally, the incidence, treatment, and days missed as a result of injury were assessed.

Results: Over 15 consecutive seasons, there were a total of 211 PM injuries. Of these injuries, 132 were classified as strains and 79 as ruptures. The incidence of strains was 0.41 per 10,000 athlete-exposures, compared with 0.25 per 10,0000 athlete-exposures for ruptures ($P < 0.01$). Players with PM ruptures treated operatively missed significantly more days than players treated nonoperatively (146.7 ± 55.0 vs 77.2 ± 72.9; $P < 0.01$).

Conclusion: NFL athletes miss significantly more time after operative compared with nonoperative management of PM ruptures.

Clinical Relevance: PM injuries are rare, with the current literature lacking description of these injuries in NFL athletes. The paucity of data limits physicians from providing adequate counseling and expectations for athletes with this injury. This research represents the largest study assessing PM injuries in NFL athletes.

Keywords: pectoralis major sprains; pectoralis major ruptures; National Football League

Pectoralis major (PM) injuries are relatively rare and primarily occur in males during vigorous athletic activity.[2,6,9,22,23] However, their frequency is rising given increased participation in sports, weight lifting, and active lifestyles in both the athletic and the general population.[6,8,11,14,20]

The PM muscle has 2 segments representing its 2 origins, the clavicular and sternal heads, which join to insert at the lateral lip of the bicipital groove of the humerus.[1,20] The muscle primarily functions to internally rotate, flex, and adduct the arm, but also contributes to shoulder stabilization. The PM muscle can rupture anywhere from its origin to insertion, and injuries result from both direct and indirect mechanisms. Direct trauma typically results in muscle belly injuries, while indirect mechanisms cause injury at the humeral insertion or musculotendinous junction.[9,15,18,26] Excessive eccentric contractions produce a majority of injuries sustained from an indirect mechanism.[19,22] Given their relatively infrequent nature, there is a lack of information about PM injuries, particularly with regard to high-risk competitive athletes.

Football players are often identified as an at-risk population for PM tendon tears given the demands of the sport.[5,16,22,26] Previous investigations are limited by small sample sizes and

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focus solely on grade 3 injuries.[10,23] In football, the blocking position with the arm abducted, externally rotated, and extended places the PM under maximum tension, with a subsequent eccentric contraction placing these athletes at risk of injury.[9] Alternatively, severe force applied to a maximally contracted muscle when attempting to brace for a fall may result in injury.[9] When injuries occur, they have a high impact on players and teams, regularly leading to significant loss of time from activity.

The aims of this study were to determine the incidence of PM injuries in National Football League (NFL) athletes and compare time missed for strains and ruptures treated with nonoperative versus operative management. Furthermore, the data were analyzed by season, session, position, activity, and contact type at the time of injury. We hypothesize that there will be a low incidence of PM injuries in NFL athletes, with athletes missing more time after operative compared with conservative management.

METHODS

After approval from the NFL Injury and Safety Committee and our institutional review board, we retrospectively reviewed reported PM injuries documented in the NFL Injury Surveillance System (NFLISS) database over a 15-season period, from 2000 through 2014. This database was established in 1980 to more consistently monitor and examine injuries sustained by players. Club staff collects data prospectively on injury type, severity, onset of injury, player return-to-participation status, position, activity during which the injury occurred, and other factors relating to playing conditions. Data are then collected and analyzed by the Quintiles Injury Surveillance and Analytics team. In the 2014 NFL season, the clinical electronic medical record (EMR) system was launched among all clubs, accompanied by the transition of injury data collection from the Quintiles Injury Surveillance System to surveillance data derived directly from the NFL EMR.

The NFL EMR collects more comprehensive injury data, resulting in more injuries being captured. Additional considerations when examining trends in injury occurrence over time include changes in the game and injury reporting. Changes in the game include rule and roster size changes, variation in number of practices and games per season, and variation in the total number of plays per game over time, which gives players a greater on-field "exposure" to the chance of being injured. Changes in injury reporting potentially result in under-ascertainment of injury data in earlier years and include greater emphasis on injury awareness and timely reporting; improvements in the injury collection system, including updates to injury codes; and transition from surveillance system to clinical EMR in the 2014 season.

The NFL records all reportable injuries through the NFLISS. A reportable injury includes all injuries resulting in time loss, defined as missing any practices or games, including being removed from the remainder of the practice or game in which the injury occurred. Players may experience more than 1 injury at a time, and in such cases, athletic trainers must designate a single injury as the "primary" injury based on severity, with additional injuries reported as secondary, tertiary, or quaternary.

In this search, PM injuries were defined as any sprain, partial tear, or complete rupture of the PM muscle or tendon. A 3-point classification was used to classify injuries: grade 1, contusion or sprain; grade 2, partial tear; and grade 3, complete rupture.[24] For this study, injuries were classified into 1 of 2 categories: PM strains (grade 1 or 2 injuries) and PM ruptures (grade 3 injuries).

The incidence of PM strains and ruptures per 10,000 athlete-exposures was evaluated. An athlete-exposure is defined as a single NFL athlete participating in a single practice or game.[13] Athlete-exposures were calculated based on possible participation for any length of time in a practice or game, assuming 53 players per team and 105 practices per year. The analysis accounts for change in league size (31 NFL teams from 2000-2001, 32 NFL teams from 2002-2014) and fewer athlete-exposures in 2011 due to cancellation of the preseason Hall of Fame game. Game-related information was derived from the NFL Game Statistics and Information System (http://www.nflgsis.com). These assumptions are limited in that fewer than 53 players may be exposed to this injury in each game or practice due to lack of playing time and other football-related factors.

The NFLISS database was used to search all primary, secondary, tertiary, and quaternary PM strains and ruptures that occurred from 2000 to 2014. These injuries were recorded by year of injury, time of season (preseason, regular season, or postseason), session (practice or game) in which they occurred, player position, activity and contact type at time of injury, management (operative vs nonoperative), and total days missed due to injury.

Prior to the 2014 season, injuries that occurred during offseason or conditioning activities were not mandatory to report to the surveillance system. Therefore, these injuries were excluded from this analysis to provide a more accurate comparison across years. Player position, activity, and contact type at the time of injury is not consistently captured when the injury occurs during a practice session. Therefore, only game-related injuries were included for these analyses. The data collection system only associates surgery and data regarding days missed with the primary injury; therefore, all secondary, tertiary, and quaternary injuries were excluded from this analysis. Days missed was defined as the number of days missed until the player returned to full participation from injury, and was calculated using the onset of injury date and return to full participation date.

All descriptive and inferential statistics were computed using R version 3.3.2 (The R Foundation for Statistical Computing, https://www.r-project.org/). Comparisons were made between the number of injuries during games, during practice, and overall for each type of injury: PM strain and PM rupture. For in-game injuries only, the number of injuries sustained by players on offense was compared with the number sustained by players on defense and the number sustained by players on special teams. Also, the number of strains was compared with

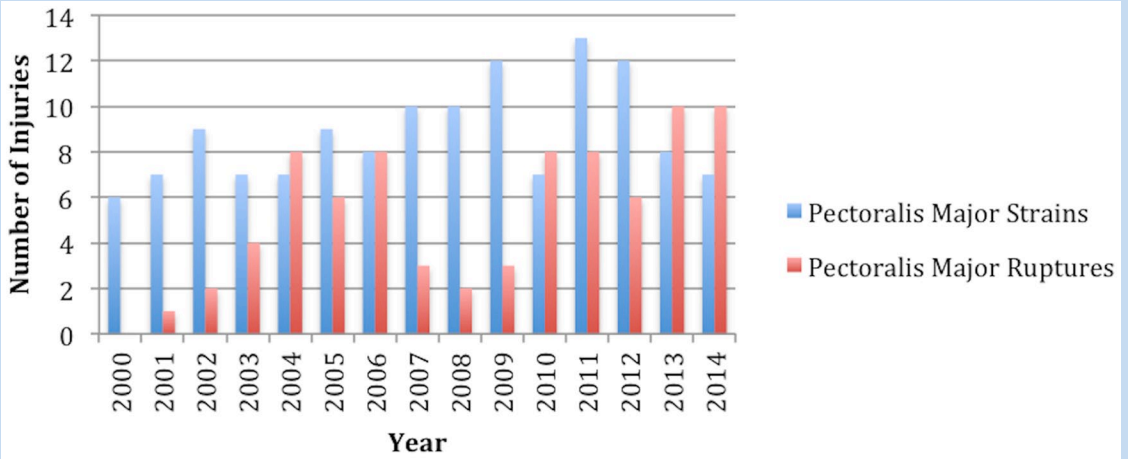

Figure 1. Pectoralis major injuries by year. Because of changes in injury reporting over time, injury counts may not be comparable across years.

the number of ruptures for games, practices, and overall as well as for offense, defense, and special teams for in-game injuries. A 2-sided exact Poisson test was used for these comparisons by testing the ratio of incidence rates (number of injuries per 10,000 athlete-exposures). For players who sustained a PM injury, 2-tailed *t* tests were used to compare the average number of games missed for athletes who did and did not require surgery as well as between strains and ruptures. Games missed for strains (managed with and without surgery) were also compared with games missed for ruptures managed with and without surgery.

RESULTS

Over 15 consecutive NFL seasons (2000-2014), there were a total of 211 PM injuries reported in the NFLISS database. Of these injuries, 132 were classified as PM strains and 79 were classified as PM ruptures (Figure 1).

Incidence of PM Injuries

The incidence of PM strain was 0.41 per 10,000 athlete-exposures. Analysis by session at time of injury resulted in a significantly larger incidence of strains during a game compared with during practice ($P < 0.01$). The incidence of PM rupture was 0.25 per 10,000 athlete-exposures. Similar to PM strains, ruptures were found to have a significantly larger incidence during a game compared with during practice ($P < 0.01$). There was a significantly higher incidence of strains compared with ruptures ($P < 0.01$) (Table 1).

PM Injuries by Season and Session

Of the 132 PM strains, a majority occurred during the regular season (68.94%), specifically during in-game activities. Similar results were seen for the 79 PM ruptures analyzed. Ruptures frequently occurred during the regular season (72.15%), specifically during in-game activities (Table 2).

PM Injuries by Position, Activity, and Contact Type

There were 101 in-game PM strains and 70 in-game PM ruptures reported. For both strains and ruptures, defensive players were significantly more likely to suffer an injury than were offensive players ($P < 0.01$) or special teams players ($P < 0.01$) (Table 3).

For the 101 in-game PM strains, the most common activity at time of injury was during a pass play (36.63%), followed by a run play (35.64%). In 70 in-game PM ruptures, 27 injuries (38.57%) were during a run play (Table 4).

Of the 101 PM strains, the most common contact type at the time of injury was contact with another player (85.15%), followed by tackling (51.49%), and blocking (20.79%). In 70 in-game PM ruptures, the frequency of contact type at the time of injury was similar to that seen after a PM strain (Table 5).

Treatment and Return to Play After PM Injuries

Treatment and return-to-play data were available for 130 players with PM strains, with 10 injuries requiring surgery. The mean number of days missed for all players with PM strains was 28.3 ± 48.0 days. Players who required surgery missed significantly more days on average than players who did not require surgery or where treatment was unknown ($P < 0.01$). Treatment and return-to-play data were available for 69 players with PM ruptures, with 52 injuries requiring surgery. The mean number of days missed for all players with PM ruptures was 129.6 ± 66.5 days. Players who required surgery missed significantly more days on average than players who did not require surgery or where treatment was unknown ($P < 0.01$) (Table 6).

Comparison between cohorts demonstrated that players who sustained PM strains missed significantly fewer games than players who sustained PM ruptures ($P < 0.01$). There was no significant difference in mean games missed follow surgery for PM strains versus PM ruptures ($P = 0.84$). Players treated

Table 1. Incidence of pectoralis major injuries

Injury and Session at Time of Injury	Number of Injuries	Athlete-Exposures	Incidence per 10,000 Athlete-Exposures	*P* (Strain vs Rupture)
Pectoralis major strain	132	3,187,950	0.41	
Practice	31	2,660,070	0.12	
Game	101	527,880	1.91	
P (game vs practice)			<0.01	
Pectoralis major rupture	79	3,187,950	0.25	<0.01
Practice	9	2,660,070	0.03	<0.01
Game	70	527,880	1.33	0.02
P (game vs practice)			<0.01	

Table 2. Pectoralis major injuries by season and session

Season and Session at Time of Injury	Number of Injuries (% of Total)
Total pectoralis major strains	132
Strains by season	
Preseason	40 (30.30)
Regular season	91 (68.94)
Postseason	1 (0.76)
Strains by session	
Practice	31 (23.48)
Game	101 (76.52)
Total pectoralis major ruptures	79
Ruptures by season	
Preseason	20 (25.32)
Regular season	57 (72.15)
Postseason	2 (2.53)
Ruptures by session	
Practice	9 (11.39)
Game	70 (88.61)

without surgery or where surgical status was unknown after a strain missed significantly fewer games than players after a rupture ($P < 0.01$).

DISCUSSION

PM injuries are uncommon in the general population, with a majority of injuries sustained in males during athletic activity.[2,12,25] In a meta-analysis of 112 cases of PM rupture, these injuries were most common during sporting activity, including weight training, weight lifting, wrestling, bodybuilding, water skiing, and football.[2,6,19,22] PM injury occurs via direct or indirect mechanisms, with indirect being more common as the muscle undergoes forced abduction against resistance, involuntary contraction, or severe traction of the arm is applied.[19]

Mechanically speaking, most PM injuries occur when maximal force is applied with the PM eccentrically contracting during an abducted and externally rotated shoulder movement.[6]

The harsh physical demands of professional football places these athletes at risk of PM muscle injury. In addition to rigorous strength training, NFL players are at risk as bracing for a fall, tackling, and blocking put the PM muscle in a vulnerable position for injury. While relatively rare, when PM injuries do occur, they have a substantial impact on teams and players, inevitably resulting in missed time from sport.

The incidence of PM strains in this study was 0.41 per 10,000 athlete-exposures, compared with 0.25 per 10,000 athlete-exposures for PM ruptures. This injury has a low incidence in NFL athletes. Of 1534 NFL quarterbacks, PM injuries comprised only 1.2% of all injuries to the glenohumeral joint and upper arm.[10] A total of 10 ruptures were reported during an 11-year study period when looking at PM ruptures in NFL athletes, with an incidence of 0.004.[23] It is unclear why inconsistency in data sets exists. However, previous investigators have postulated that underreporting of PM ruptures occurs because there was no specific ICD-9 (International Classification of Diseases, Ninth Revision) code for PM tendon rupture; rather, this injury was part of a larger grouping for upper extremity tendon injury.[3]

Table 3. In-game pectoralis major injuries by position

Position at Time of Injury	Number of Injuries (% of Total)	*P* (Strain vs Rupture)
Total in-game pectoralis major strains	101	
Offense	32 (31.68)	
Offensive line	17 (16.83)	
Running back	4 (3.96)	
Tight end	6 (5.94)	
Quarterback	3 (2.97)	
Wide receiver	2 (1.98)	
Defense	60 (59.41)	
Defensive secondary	15 (14.85)	
Defensive line	25 (24.75)	
Linebacker	20 (19.80)	
Special teams	9 (8.91)	
P		
Defense vs offense	<0.01	
Offense vs special teams	<0.01	
Defense vs special teams	<0.01	
Total in-game pectoralis major ruptures	70	
Offense	12 (17.14)	<0.01
Offensive line	7 (10.00)	
Running back	3 (4.29)	
Tight end	1 (1.43)	
Quarterback	1 (1.43)	
Wide receiver	0 (0.00)	
Defense	42 (60.00)	0.09
Defensive secondary	16 (22.86)	
Defensive line	10 (14.29)	
Linebacker	16 (22.86)	
Special teams	16 (22.86)	0.23
P		
Defense vs offense	<0.01	
Offense vs special teams	0.57	
Defense vs special teams	<0.01	

Table 4. In-game pectoralis major injuries by activity

Activity at Time of Injury	Number of Injuries (% of Total)
Total in-game pectoralis major strains	101
Pass play	37 (36.63)
Run play	36 (35.64)
Fumble/interception	2 (1.98)
Kickoff return	6 (5.94)
Punt return	2 (1.98)
Field goal attempt	1 (0.99)
Other	17 (16.83)
Total in-game pectoralis major ruptures	70
Pass play	13 (18.57)
Run play	27 (38.57)
Fumble/interception	1 (1.43)
Kickoff return	8 (11.43)
Punt return	6 (8.57)
Field goal attempt	1 (1.43)
Other	14 (20.00)

Table 5. In-game pectoralis major injuries by contact type

Contact Type	Number of Injuries (% of Total)
Total in-game pectoralis major strains	101
Contact with another player	86 (85.15)
Tackling	52 (51.49)
Tackled	5 (4.95)
Blocking	21 (20.79)
Blocked	8 (7.92)
Fighting/horseplay	0 (0.00)
Contact with other	5 (4.95)
Noncontact	4 (3.96)
Unknown	6 (5.94)
Total in-game pectoralis major ruptures	70
Contact with another player	58 (82.86)
Tackling	48 (68.57)
Tackled	1 (1.43)
Blocking	7 (10.00)
Blocked	1 (1.43)
Fighting/horseplay	1 (1.43)
Contact with other	6 (8.57)
Noncontact	0 (0.00)
Unknown	6 (8.57)

In this study, PM injuries most frequently occurred during the regular season, specifically during in-game activity, which is consistent with previous reports.[23] In-game injury occurs more frequently given the unpredictable nature of impact during these sessions as compared with practice; particularly, as it appears, an indirect mechanism is primarily responsible for the majority of injuries. Contact with another player, which is the primary cause of injury, during practice is decreased after limitations on full-padded, full-contact practices during the regular season.

Contact with another player while tackling and blocking resulted in the most injuries in both cohorts. This is consistent with a previous study reporting tackling to be associated with the most PM ruptures.[23] Again, given that an indirect mechanism resulting in tensioning of the PM tendon is principally responsible for injury, it is intuitive that abrupt contact during tackling leads to a majority of these injuries. The correlation between PM injury and defensive players is higher given that the primary objective on defense is tackling versus the primary objective on offense is blocking.

As expected, the mean days missed was significantly less for PM strains (28.3 ± 48.0 days) than for PM ruptures (129.6 ± 66.5 days) ($P < 0.01$). Athletes in each cohort who underwent surgery missed more days than those treated without surgery or where the surgical status was unknown. There was no significant difference between mean days missed between the 2 cohorts when treated surgically ($P = 0.84$). Only 7.69% of athletes with a PM strain required surgery compared with 75.36% of athletes with a PM rupture. Given the nature of the deidentified data set, surgical indications for those undergoing operative intervention in the PM strain cohort are unknown. Surgical intervention for complete ruptures is appropriate, as multiple studies have demonstrated superior outcomes after surgical repair of PM ruptures than when treated conservatively.[1,2,7,21,22,25] Patients with PM ruptures treated conservatively have marked deficit in peak torque and work/repetition using the isokinetic testing; moreover, conservatively treated athletes have difficulty returning to their previous level of athletic functional ability.[25,26] The best outcomes have been reported in ruptures repaired acutely, compared with conservative or delayed repair.[1,2,15,17] While both conservative and surgical management allow for restoration of full range of

Table 6. Treatment and return to play after pectoralis major injury

Treatment	Number of Injuries (%)	Mean ± SD Days Missed (Range)	*P* (Strains vs Ruptures)
Pectoralis major strains	130	28.3 ± 48.0 (0-237)	
Required surgery	10 (7.69)	151.2 ± 62.9 (8-237)	
No surgery/unknown	120 (92.31)	18.1 ± 28.7 (0-228)	
P (surgery vs no surgery)		<0.01	
Pectoralis major ruptures	69	129.6 ± 66.5 (2-260)	<0.01
Required surgery	52 (75.36)	146.7 ± 55.0 (9-252)	0.84
No surgery/unknown	17 (24.64)	77.2 ± 72.9 (2-260)	<0.01
P (surgery vs no surgery)		<0.01	

motion and pain relief, operative intervention appears to be essential for restoration of normal strength and recovery of peak torques and work performed, especially for the athlete.[2,4,12]

Limitations

There are several limitations in this study. The NFLISS is a well-established reporting system; however, it relies on clubs to consistently and accurately report injury data. It is possible that injuries were unreported by the medical teams. Furthermore, changes in the game and injury reporting complicate examination of injury trends over time, with changes in injury reporting potentially resulting in under-ascertainment of injury data in earlier years. Days missed was calculated using the onset of injury date and return to full participation date. Athletic trainers can either enter in the actual date of return to play date or an estimated date in cases where the actual return date is unknown or the player returned in the offseason. Limitations of this variable include the following: return dates do not always reflect the exact date of return, which may result in calculations of days missed to be slightly lower or higher than the true number of days missed for the injury; and injuries with unknown or missing return-to-play dates are excluded from calculations due to a player's not yet having returned or data entries not being completed by the data cutoff (June 22, 2015). This may be differential based on timing in season or severity of injury, and time-to-return statistics may show an artificially low amount of time lost in cases where a more severe injury kept a player out past June 2015 for the 2014 season or earlier. Injuries that occurred during offseason or conditioning activities, such as weight lifting, were not mandatory to report; thus, these injuries were excluded from this analysis to provide a more accurate comparison across years. Apart from data collection, another limitation of our study is the lack of outcome data to compare player performance after operative and nonoperative management. However, given that the data are anonymous to protect the confidentiality of these professional athletes, it was impossible to collect outcomes data.

CONCLUSION

Athletes with PM ruptures treated operatively miss significantly more days than players treated nonoperatively. With timely and appropriate treatment, athletes are able to return to competitive play.

Clinical Recommendations

SORT: Strength of Recommendation Taxonomy
A: consistent, good-quality patient-oriented evidence
B: inconsistent or limited-quality patient-oriented evidence
C: consensus, disease-oriented evidence, usual practice, expert opinion, or case series

Clinical Recommendation	SORT Evidence Rating
Pectoralis major injuries are rare in National Football League athletes	C
Athletes should be counseled that pectoralis major injuries requiring surgical intervention result in significantly more time missed compared with athletes with pectoralis major injuries treated nonoperatively	C

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